

Unit-1 Introduction to AI:**GTU Syllabus:**

- History & overview of Artificial Intelligence
- Definition of Artificial Intelligence
- Artificial Narrow Intelligence, Artificial General Intelligence, Artificial Super Intelligence
- Concepts of Production, Agents and Environments
- Characteristic of Intelligent Agents, Concept of Rationality, Nature of Environments.

What is Artificial Intelligence (AI)?

In today's world, technology is growing very fast, and we are getting in touch with different new technologies day by day.

Here, one of the booming technologies of computer science is Artificial Intelligence which is ready to create a new revolution in the world by making intelligent machines. The Artificial Intelligence is now all around us. It is currently working with a variety of subfields, ranging from general to specific, such as self-driving cars, playing chess, proving theorems, playing music, Painting, etc.

AI is one of the fascinating and universal fields of Computer science which has a great scope in future. AI holds a tendency to cause a machine to work as a human.

Artificial Intelligence is composed of two words **Artificial** and **Intelligence**, where Artificial defines "*man-made*," and intelligence defines "*thinking power*", hence AI means "*a man-made thinking power*."

Definition of AI:

"It is a branch of computer science by which we can create intelligent machines which can behave like a human, think like humans, and able to make decisions."

Artificial Intelligence exists when a machine can have human based skills such as learning, reasoning, and solving problems

With Artificial Intelligence you do not need to pre program a machine to do some work, despite that you can create a machine with programmed algorithms which can work with own intelligence, and that is the awesomeness of AI.

It is believed that AI is not a new technology, and some people says that as per Greek myth, there were Mechanical men in early days which can work and behave like humans.

Why Artificial Intelligence?

Following are some main reasons to learn about AI:

- With the help of AI, you can create such software or devices which can solve real-world problems very easily and with accuracy such as health issues, marketing, traffic issues, etc.
- With the help of AI, you can create your personal virtual Assistant, such as Cortana, Google Assistant, Siri, etc.
- With the help of AI, you can build such Robots which can work in an environment where survival of humans can be at risk.
- AI opens a path for other new technologies, new devices, and new Opportunities.

Goals of Artificial Intelligence

Following are the main goals of Artificial Intelligence:

1. Replicate human intelligence
2. Solve Knowledge-intensive tasks
3. An intelligent connection of perception and action
4. Building a machine which can perform tasks that requires human intelligence such as:
 - Proving a theorem
 - Playing chess
 - Plan some surgical operation
 - Driving a car in traffic

5. Creating some system which can exhibit intelligent behavior, learn new things by itself, demonstrate, explain, and can advise to its user.

History of AI

- Throughout history, people have been intrigued by the idea of making non-living things smart. In ancient times, Greek stories mentioned gods creating clever machines, and in Egypt, engineers made statues move. Thinkers like **Aristotle and Ramon Llull** laid the groundwork for AI by describing how human thinking works using symbols.
- In the late 1800s and early 1900s, modern computing started to take shape. Charles Babbage and Ada Lovelace designed machines that could be programmed in the 1830s. In the 1940s, John Von Neumann came up with the idea of storing computer programs. At the same time, Warren McCulloch and Walter Pitts started building the basics of neural networks.
- The **1950s** brought us modern computers, letting scientists dig into machine intelligence. **Alan Turing's** Turing test became a big deal in computer smarts. The term "**artificial intelligence**" was first used in a **1956 Dartmouth College meeting**, where they introduced the first AI program, the Logic Theorist.
- The following years had good times and bad times for AI, called "**AI Winters**." In the 1970s and 1980s, we hit limits with computer power and complexity. But in the late 1990s, things got exciting again. Computers were faster, and there was more data. **IBM's Deep Blue** beating chess champion Garry Kasparov in 1997 was a big moment.
- The 2000s started a new era with machine learning, language processing, and computer vision. This led to cool new products and services. The 2010s saw AI take off with things like voice assistants and self-driving cars. Generative AI, which makes creative stuff, also started getting big.
- In the 2020s, generative AI like **ChatGPT-3** and **Google's Bard** grabbed everyone's attention. These models can create all sorts of new things when you give them a prompt, like essays or art. But remember, this tech is still new, and there are things to fix, like making sure it doesn't make things up.

Types of Artificial Intelligence:

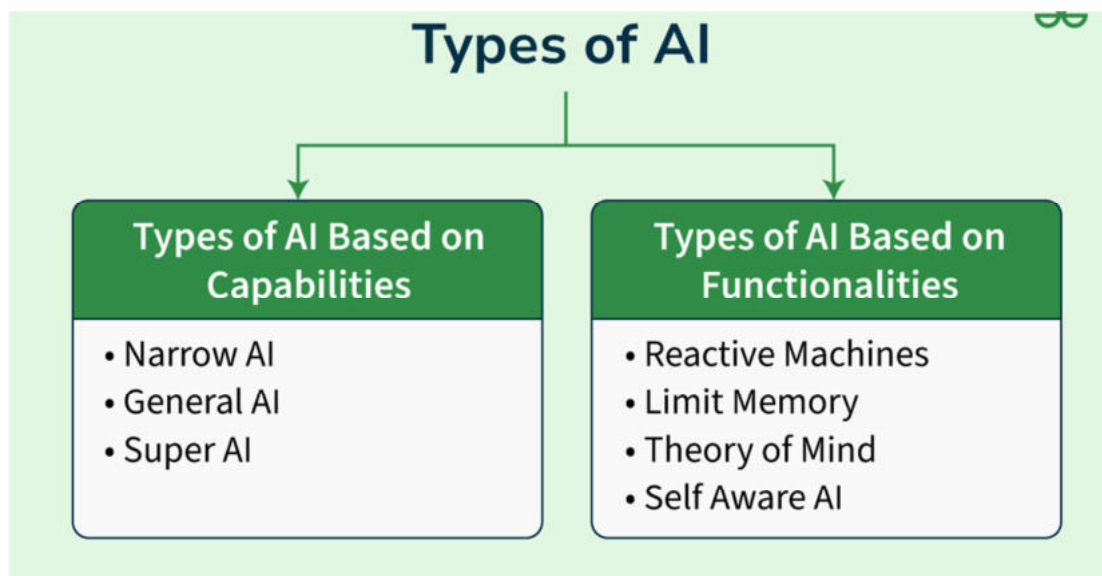
Artificial Intelligence (AI) has transformed industries, leading to significant advancements in technology, science, and everyday life. To understand AI better, we must first recognize that AI can be categorized into different types based on capabilities and functionalities.

Type 1: Based on Capabilities of AI

- Narrow AI
- General AI
- Super AI

Type 2: Based on the Functionality of AI

- Reactive Machines
- Limited Memory AI
- Theory of Mind
- Self-Aware AI



Types of AI Based on Capabilities

1. Narrow AI (Weak AI)

Narrow AI is designed and trained on a specific task or a narrow range tasks. These Narrow AI systems are designed and trained for a purpose. These Narrow

systems perform their designated tasks but mainly lack in the ability to generalize tasks.

Examples:

- Voice assistants like **Siri** or **Alexa** that understand specific commands.
- **Facial recognition** software used in security systems.
- **Recommendation engines** used by platforms like Netflix or Amazon.

Despite being highly efficient at specific tasks, Narrow AI lacks the ability to function beyond its predefined scope. These systems do not possess understanding or awareness.

2. General AI (Strong AI)

General AI refers to AI systems that have human intelligence and abilities to perform various tasks. Systems have capability to understand, learn and apply across a wide range of tasks that are similar to how a human can adapt to various tasks.

While General AI remains a theoretical concept, researchers aim to develop AI systems that can perform any intellectual task a human can. It requires the machine to have consciousness, self-awareness, and the ability to make independent decisions, which is not yet achievable.

Potential Applications:

- Robots that can learn new skills and adapt to unforeseen challenges in real-time.
- AI systems that could autonomously diagnose and solve complex medical issues across various specializations.

3. Super intelligence (Super AI)

Super AI surpasses intelligence of human in solving-problem, creativity, and overall abilities. Super AI develops emotions, desires, need and beliefs of their own. They are able to make decisions of their own and solve problem of its own. Such AI would not only be able to complete tasks better than humans but also understand and interpret emotions and respond in a human-like manner.

While **Super AI** remains speculative, it could revolutionize industries, scientific research, and problem-solving, possibly leading to unprecedented advancements. However, it also raises ethical concerns regarding control and regulation.

Types of Artificial Intelligence Based on Functionalities:

AI can also be classified into four types based on how the systems function. This classification is more commonly used to distinguish AI systems in practical applications.

1. Reactive Machines

Reactive machines are the most basic form of AI. They operate purely based on the present data and do not store any previous experiences or learn from past actions. These systems respond to specific inputs with fixed outputs and are unable to adapt.

Examples:

- **IBM's Deep Blue**, which defeated the world chess champion Garry Kasparov in 1997. It could identify the pieces on the board and make predictions but could not store any memories or learn from past games.
- **Google's AlphaGo**, which played the board game Go using a similar approach of pattern recognition without learning from previous games.

2. Limited Memory in AI

Limited Memory AI can learn from past data to improve future responses. Most modern AI applications fall under this category. These systems use historical data to make decisions and predictions but do not have long-term memory. Machine learning models, particularly in autonomous systems and robotics, often rely on limited memory to perform better.

Examples:

- **Self-driving cars:** They observe the road, traffic signs, and movement of nearby cars, and make decisions based on past experiences and current conditions.
- **Chatbots** that can remember recent conversations to improve the flow and relevance of replies.

3. Theory of Mind

Theory of Mind AI aims to understand human emotions, beliefs, intentions, and desires. While this type of AI remains in development, it would allow machines to engage in more sophisticated interactions by perceiving emotions and adjusting behavior accordingly.

Potential Applications:

- **Human-robot interaction** where AI could detect emotions and adjust its responses to empathize with humans.
- **Collaborative robots** that work alongside humans in fields like healthcare, adapting their tasks based on the needs of the patients.

4. Self-Awareness AI

Self-Aware AI is an advanced stage of AI that possesses self-consciousness and awareness. This type of AI would have the ability to not only understand and react to emotions but also have its own consciousness, similar to human awareness.

While we are far from achieving self-aware AI, it remains the ultimate goal for AI development. It opens philosophical debates about consciousness, identity, and the rights of AI systems if they ever reach this level.

Potential Applications:

- Fully autonomous systems that can make moral and ethical decisions.
- AI systems that can independently pursue goals based on their understanding of the world around them.

Concepts of Production, Agents and Environments:

Production System in AI

In artificial intelligence (AI), a production system refers to a type of rule-based system that is designed to provide a structured approach to problem solving and decision-making. This framework is particularly influential in the realm of expert systems, where it simulates human decision-making processes using a set of predefined rules and facts.

Let's consider an example of **Expert System for Medical Diagnosis**.

Scenario: A patient comes to a healthcare facility with the following symptoms: fever, severe headache, sensitivity to light, and stiff neck.

MediDiagnose operates in the following manner:

1. **Input:** A healthcare professional inputs the symptoms into MediDiagnose.
2. **Processing:**
 - **MediDiagnose** reviews its knowledge base for rules that match the given symptoms.
 - It identifies several potential conditions but recognizes a strong match for meningitis based on the combination of symptoms.
3. **Output:**
 - The system suggests that meningitis could be a possible diagnosis and recommends further tests to confirm, such as a lumbar puncture.
 - It also provides a list of other less likely conditions based on the symptoms for comprehensive differential diagnosis.

MediDiagnose uses its rule-based system to quickly filter through vast amounts of medical data to provide preliminary diagnoses. This assists doctors in focusing their investigative efforts more efficiently and potentially speeds up the process of reaching an accurate diagnosis.

Key Components of a Production System in AI

The key components of production system includes:

1. **Knowledge Base:** This is the core repository where all the rules and facts are stored. In AI, the knowledge base is critical as it contains the domain-specific information and the if-then rules that dictate how decisions are made or actions are taken.
2. **Inference Engine:** The inference engine is the mechanism that applies the rules to the known facts to derive new facts or to make decisions. It scans the rules and decides which ones are applicable based on the current facts in the working memory. It can operate in two modes:
 - **Forward Chaining (Data-driven):** This method starts with the available data and uses the inference rules to extract more data until a goal is reached.
 - **Backward Chaining (Goal-driven):** This approach starts with a list of goals and works backwards to determine what data is required to achieve those goals.
3. **Working Memory:** Sometimes referred to as the fact list, working memory holds the dynamic information that changes as the system operates. It represents the current state of knowledge, including facts that are initially known and those that are deduced throughout the operation of the system.
4. **Control Mechanism:** This governs the order in which rules are applied by the inference engine and manages the flow of the process. It ensures that the system responds appropriately to changes in the working memory and applies rules effectively to reach conclusions or solutions.

Types of Production Systems

Production systems in AI can be categorized based on how they handle and process knowledge. This categorization includes Rule-Based Systems, Procedural Systems, and Declarative Systems, each possessing unique characteristics and applications.

1. Rule-Based Systems

1. Explanation of Rule-Based Reasoning

- Rule-based systems operate by applying a set of pre-defined rules to the given data to deduce new information or make decisions. These rules are generally in the form of conditional statements (if-then statements) that link conditions with actions or outcomes.

2. Examples of Rule-Based Systems in AI

- **Diagnostic Systems:** Like medical diagnosis systems that infer diseases from symptoms.
- **Fraud Detection Systems:** Used in banking and insurance, these systems analyze transaction patterns to identify potentially fraudulent activities.

2. Procedural Systems

1. Description of Procedural Knowledge

- Procedural systems utilize knowledge that describes how to perform specific tasks. This knowledge is procedural in nature, meaning it focuses on the steps or procedures required to achieve certain goals or results.

2. Applications of Procedural Systems

- **Manufacturing Control Systems:** Automate production processes by detailing step-by-step procedures to assemble parts or manage supply chains.
- **Interactive Voice Response (IVR) Systems:** Guide users through a series of steps to resolve issues or provide information, commonly used in customer service.

3. Declarative Systems

1. Understanding Declarative Knowledge

- Declarative systems are based on facts and information about what something is, rather than how to do something. These systems store knowledge that can be queried to make decisions or solve problems.

2. Instances of Declarative Systems in AI

- **Knowledge Bases in AI Assistants:** Power virtual assistants like Siri or Alexa, which retrieve information based on user queries.
- **Configuration Systems:** Used in product customization, where the system decides on product specifications based on user preferences and declarative rules about product options.

Each type of production system offers different strengths and is suitable for various applications, from straightforward rule-based decision-making to complex systems requiring intricate procedural or declarative reasoning.

How Production Systems Function?

The operation of a production system in AI follows a cyclic pattern:

- **Match:** The inference engine checks which rules are triggered based on the current facts in the working memory.
- **Select:** From the triggered rules, the system (often through the control mechanism) selects one based on a set of criteria, such as specificity, recency, or priority.
- **Execute:** The selected rule is executed, which typically modifies the facts in the working memory, either by adding new facts, changing existing ones, or removing some.

Applications of Production Systems in AI

Production systems are used across various domains where decision-making can be encapsulated into clear, logical rules:

- **Expert Systems:** For diagnosing medical conditions, offering financial advice, or making environmental assessments.
- **Automated Planning:** Used in logistics to optimize routes and schedules based on current data and objectives.
- **Game AI:** Manages non-player character behavior and decision-making in complex game environments.

Agent and Environment:

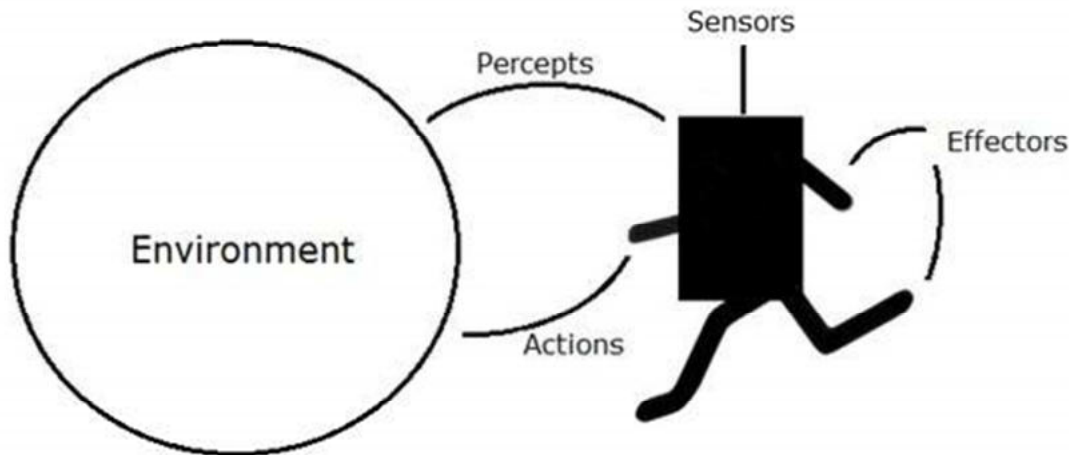
An AI system is composed of an agent and its environment. The agents act in their environment. The environment may contain other agents.

What are Agent and Environment?

An **agent** is anything that can perceive its environment through **sensors** and acts upon that environment through **effectors**.

- A **human agent** has sensory organs such as eyes, ears, nose, tongue and skin parallel to the sensors, and other organs such as hands, legs, mouth, for effectors.
- A **robotic agent** replaces cameras and infrared range finders for the sensors, and various motors and actuators for effectors.

- A **software agent** has encoded bit strings as its programs and actions.



Agent Terminology

- **Performance Measure of Agent** – It is the criteria, which determines how successful an agent is.
- **Behavior of Agent** – It is the action that agent performs after any given sequence of percepts.
- **Percept** – It is agent's perceptual inputs at a given instance.
- **Percept Sequence** – It is the history of all that an agent has perceived till date.
- **Agent Function** – It is a map from the precept sequence to an action.

Types of AI Agents:

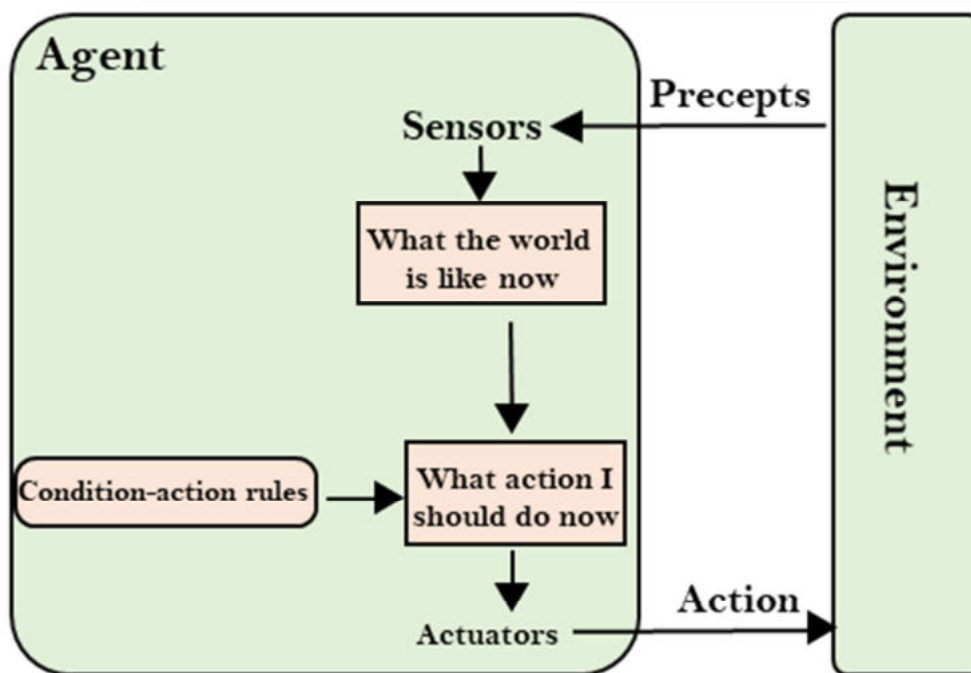
Agents can be grouped into five classes based on their degree of perceived intelligence and capability. All these agents can improve their performance and generate better action over the time. These are given below:

- Simple Reflex Agent
- Model-based reflex agent
- Goal-based agents
- Utility-based agent
- Learning agent

1. Simple Reflex agent:

- The Simple reflex agents are the simplest agents. These agents take decisions on the basis of the current percepts and ignore the rest of the percept history.

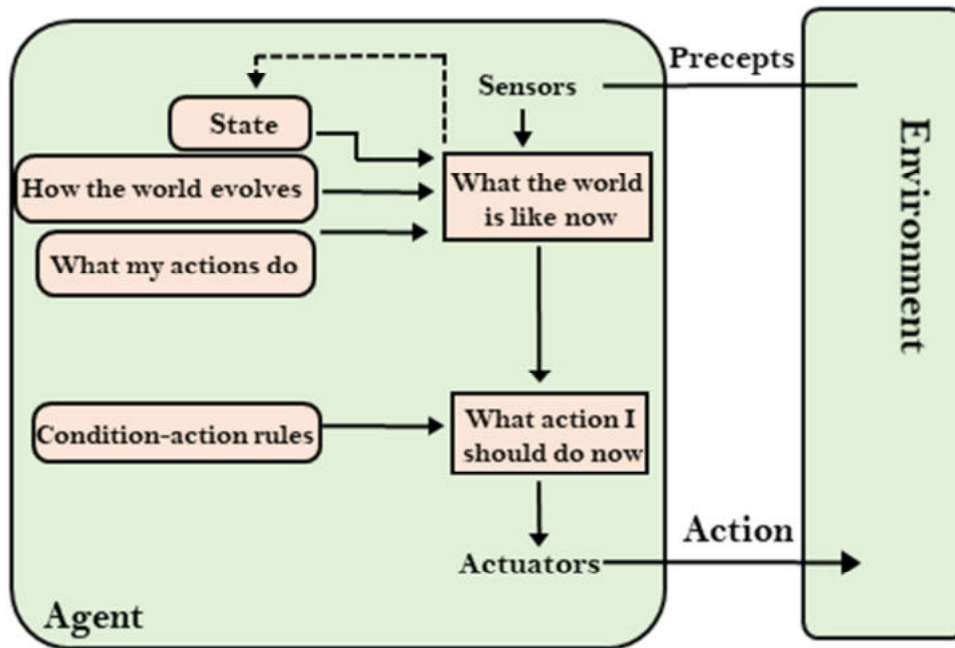
- These agents only succeed in the fully observable environment.
- The Simple reflex agent does not consider any part of percepts history during their decision and action process.
- The Simple reflex agent works on **Condition-action rule**, which means it maps the current state to action. Such as a Room Cleaner agent, it works only if there is dirt in the room.
- Problems for the simple reflex agent design approach:
 - They have very limited intelligence
 - They do not have knowledge of non-perceptual parts of the current state
 - Mostly too big to generate and to store.
 - Not adaptive to changes in the environment.



2. Model-based reflex agent

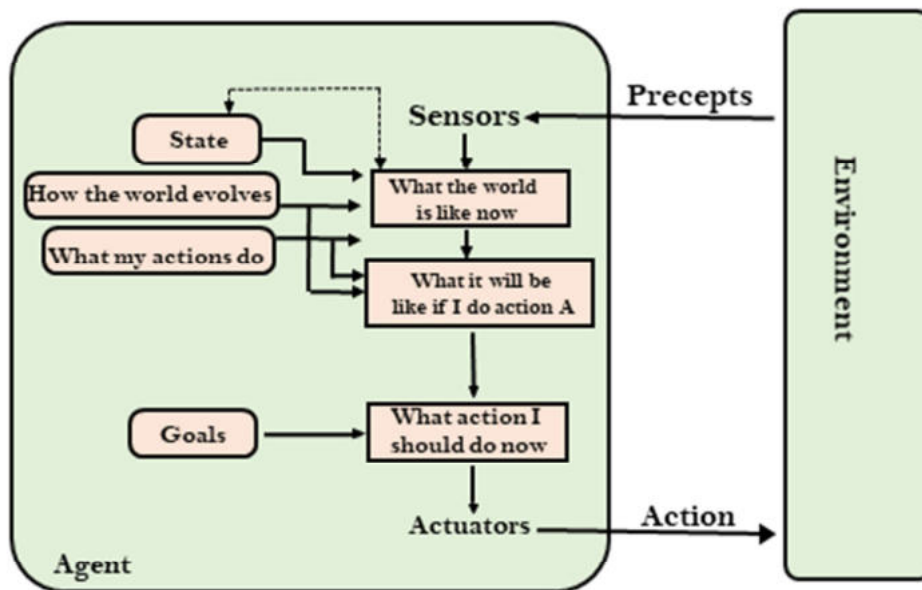
- The Model-based agent can work in a partially observable environment, and track the situation.
- A model-based agent has two important factors:
 - **Model:** It is knowledge about "how things happen in the world," so it is called a Model-based agent.
 - **Internal State:** It is a representation of the current state based on percept history.
- These agents have the model, "which is knowledge of the world" and based on the model they perform actions.
- Updating the agent state requires information about:

- How the world evolves
- How the agent's action affects the world.



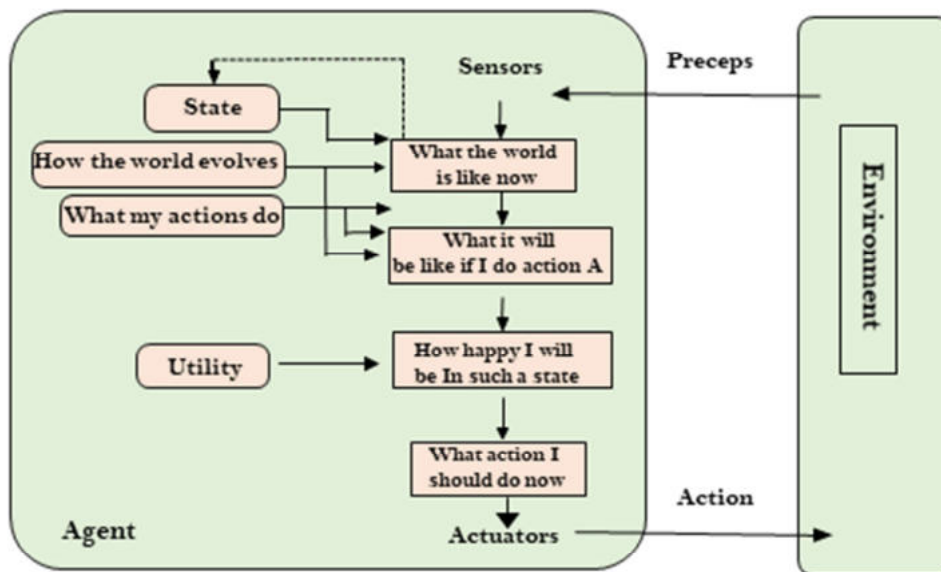
3. Goal-based agents

- The knowledge of the current state environment is not always sufficient to decide for an agent to what to do.
- The agent needs to know its goal which describes desirable situations.
- Goal-based agents expand the capabilities of the model-based agent by having the "goal" information.
- They choose an action, so that they can achieve the goal.
- These agents may have to consider a long sequence of possible actions before deciding whether the goal is achieved or not. Such considerations of different scenario are called searching and planning, which makes an agent proactive.



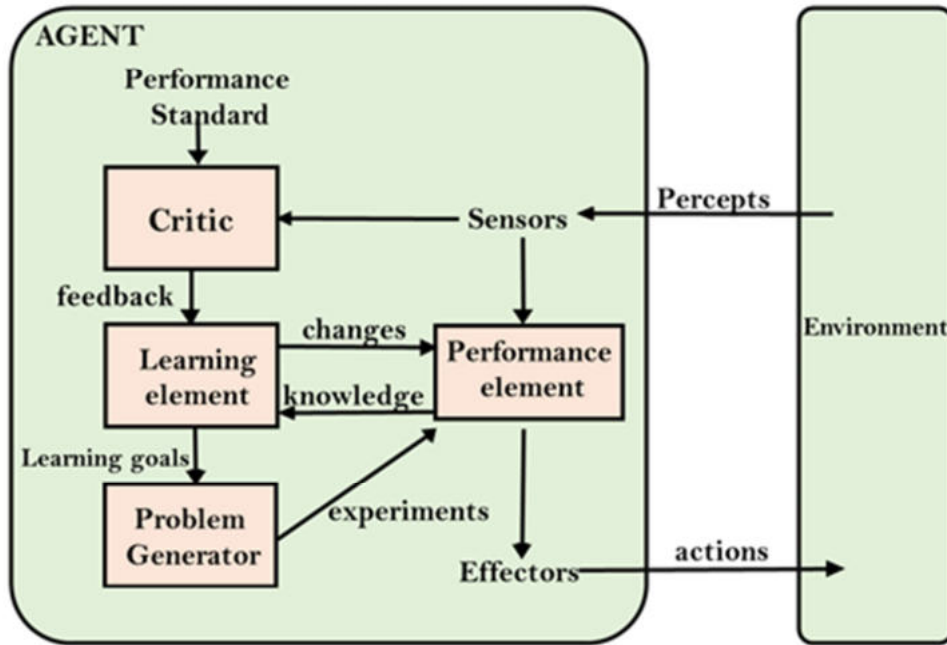
4. Utility-based agents

- These agents are similar to the goal-based agent but provide an extra component of utility measurement which makes them different by providing a measure of success at a given state.
- Utility-based agent act based not only goals but also the best way to achieve the goal.
- The Utility-based agent is useful when there are multiple possible alternatives, and an agent has to choose in order to perform the best action.
- The utility function maps each state to a real number to check how efficiently each action achieves the goals.



5. Learning Agents

- A learning agent in AI is the type of agent which can learn from its past experiences, or it has learning capabilities.
- It starts to act with basic knowledge and then able to act and adapt automatically through learning.
- A learning agent has mainly four conceptual components, which are:
 - **Learning element:** It is responsible for making improvements by learning from environment
 - **Critic:** Learning element takes feedback from critic which describes that how well the agent is doing with respect to a fixed performance standard.
 - **Performance element:** It is responsible for selecting external action
 - **Problem generator:** This component is responsible for suggesting actions that will lead to new and informative experiences.
- Hence, learning agents are able to learn, analyze performance, and look for new ways to improve the performance.



Agents in Artificial Intelligence

An AI system can be defined as the study of the rational agent and its environment. The agents sense the environment through sensors and act on their environment through actuators. An AI agent can have mental properties such as knowledge, belief, intention, etc.

What is an Agent?

An agent can be anything that perceives its environment through sensors and act upon that environment through actuators. An Agent runs in the cycle of **perceiving**, **thinking**, and **acting**. An agent can be:

- **Human-Agent:** A human agent has eyes, ears, and other organs which work for sensors and hand, legs, vocal tract work for actuators.
- **Robotic Agent:** A robotic agent can have cameras, infrared range finder, NLP for sensors and various motors for actuators.
- **Software Agent:** Software agent can have keystrokes, file contents as sensory input and act on those inputs and display output on the screen.

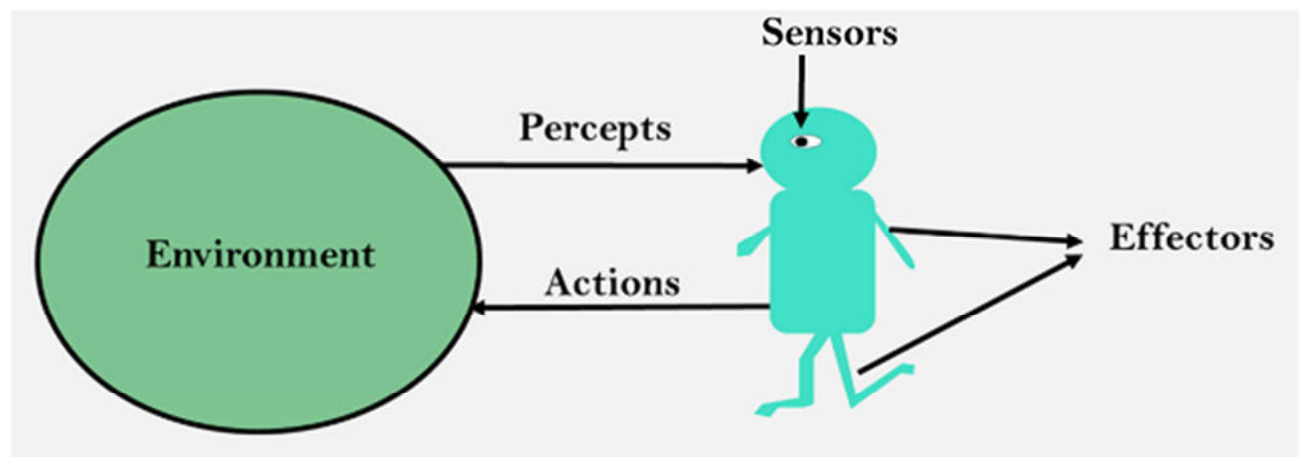
Hence the world around us is full of agents such as thermostat, cellphone, camera, and even we are also agents.

Before moving forward, we should first know about sensors, effectors, and actuators.

Sensor: Sensor is a device which detects the change in the environment and sends the information to other electronic devices. An agent observes its environment through sensors.

Actuators: Actuators are the component of machines that converts energy into motion. The actuators are only responsible for moving and controlling a system. An actuator can be an electric motor, gears, rails, etc.

Effectors: Effectors are the devices which affect the environment. Effectors can be legs, wheels, arms, fingers, wings, fins, and display screen.



Intelligent Agents:

An intelligent agent is an autonomous entity which act upon an environment using sensors and actuators for achieving goals. An intelligent agent may learn from the environment to achieve their goals. A thermostat is an example of an intelligent agent.

Following are the main four rules for an AI agent:

- **Rule 1:** An AI agent must have the ability to perceive the environment.
- **Rule 2:** The observation must be used to make decisions.
- **Rule 3:** Decision should result in an action.
- **Rule 4:** The action taken by an AI agent must be a rational action.

Rational Agent:

A rational agent is an agent which has clear preference, models uncertainty, and acts in a way to maximize its performance measure with all possible actions.

A rational agent is said to perform the right things. AI is about creating rational agents to use for game theory and decision theory for various real-world scenarios.

For an AI agent, the rational action is most important because in AI reinforcement learning algorithm, for each best possible action, agent gets the positive reward and for each wrong action, an agent gets a negative reward.

Note: Rational agents in AI are very similar to intelligent agents.

Rationality:

The rationality of an agent is measured by its performance measure. Rationality can be judged on the basis of following points:

- Performance measure which defines the success criterion.
- Agent prior knowledge of its environment.
- Best possible actions that an agent can perform.
- The sequence of percepts.

Note: Rationality differs from Omniscience because an Omniscient agent knows the actual outcome of its action and act accordingly, which is not possible in reality.

Structure of an AI Agent

The task of AI is to design an agent program which implements the agent function. The structure of an intelligent agent is a combination of architecture and agent program. It can be viewed as:

1. Agent = Architecture + Agent program

Following are the main three terms involved in the structure of an AI agent:

Architecture: Architecture is machinery that an AI agent executes on.

Agent Function: Agent function is used to map a percept to an action.

1. $f: P^* \rightarrow A$

Agent program: Agent program is an implementation of agent function. An agent program executes on the physical architecture to produce function f .

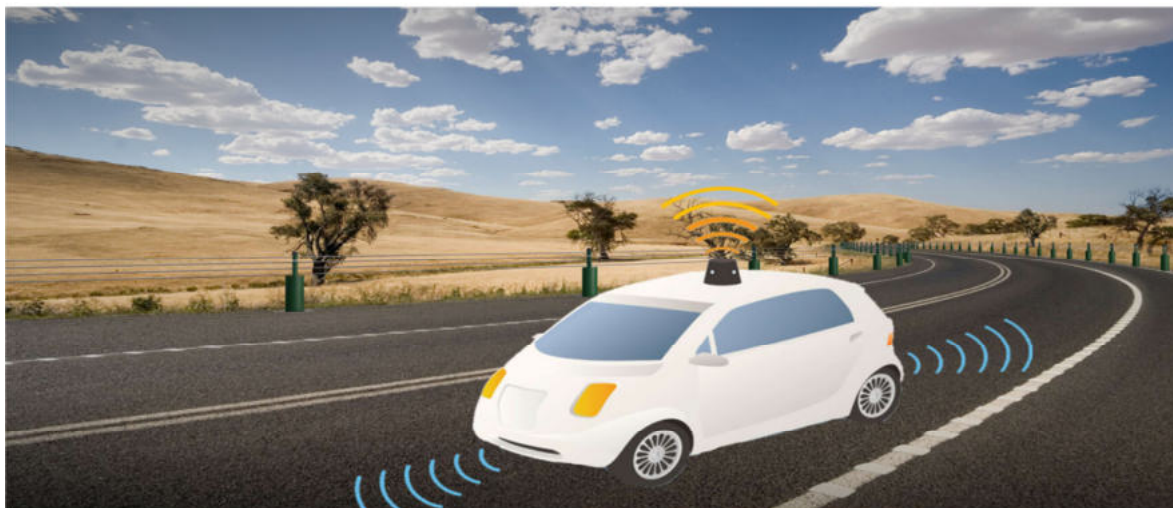
PEAS Representation

PEAS is a type of model on which an AI agent works upon. When we define an AI agent or rational agent, then we can group its properties under PEAS representation model. It is made up of four words:

- **P:** Performance measure
- **E:** Environment
- **A:** Actuators
- **S:** Sensors

Here performance measure is the objective for the success of an agent's behavior.

PEAS for self-driving cars:



Let's suppose a self-driving car then PEAS representation will be:

Performance: Safety, time, legal drive, comfort

Environment: Roads, other vehicles, road signs, pedestrian

Actuators: Steering, accelerator, brake, signal, horn

Sensors: Camera, GPS, speedometer, odometer, accelerometer, sonar.

Example of Agents with their PEAS representation

Agent	Performance measure	Environment	Actuators	Sensors
1. Medical Diagnose	○ Healthy patient ○ Minimized cost	○ Patient ○ Hospital ○ Staff	○ Tests ○ Treatments	Keyboard (Entry of symptoms)
2. Vacuum Cleaner	○ Cleanness ○ Efficiency ○ Battery life ○ Security	○ Room ○ Table ○ Wood floor ○ Carpet ○ Various obstacles	○ Wheels ○ Brushes ○ Vacuum Extractor	○ Camera ○ Dirt detection sensor ○ Cliff sensor ○ Bump Sensor ○ Infrared Wall Sensor
3. Part - picking Robot	○ Percentage of parts in correct bins.	○ Conveyor belt with parts, ○ Bins	○ Jointed Arms ○ Hand	○ Camera ○ Joint angle sensors.

Types of Environments in AI

An environment in artificial intelligence is the surrounding of the agent. The agent takes input from the environment through sensors and delivers the output to the environment through actuators. There are several types of environments:

- Fully Observable vs Partially Observable
- Deterministic vs Stochastic
- Competitive vs Collaborative
- Single-agent vs Multi-agent
- Static vs Dynamic
- Discrete vs Continuous
- Episodic vs Sequential
- Known vs Unknown

1. Fully Observable vs Partially Observable

- When an agent sensor is capable to sense or access the complete state of an agent at each point in time, it is said to be a fully observable environment else it is partially observable.
- Maintaining a fully observable environment is easy as there is no need to keep track of the history of the surrounding.
- An environment is called **unobservable** when the agent has no sensors in all environments.
- **Examples:**
 - **Chess** – the board is fully observable, and so are the opponent's moves.
 - **Driving** – the environment is partially observable because what's around the corner is not known.

2. Deterministic vs Stochastic

- When a uniqueness in the agent's current state completely determines the next state of the agent, the environment is said to be deterministic.
- The stochastic environment is random in nature which is not unique and cannot be completely determined by the agent.
- **Examples:**
 - **Chess** – there would be only a few possible moves for a chess piece at the current state and these moves can be determined.
 - **Self-Driving Cars**- the actions of a self-driving car are not unique, it varies time to time.

3. Competitive vs Collaborative

- An agent is said to be in a competitive environment when it competes against another agent to optimize the output.
- The game of chess is competitive as the agents compete with each other to win the game which is the output.
- An agent is said to be in a collaborative environment when multiple agents cooperate to produce the desired output.
- When multiple self-driving cars are found on the roads, they cooperate with each other to avoid collisions and reach their destination which is the output desired.

4. Single-agent vs Multi-agent

- An environment consisting of only one agent is said to be a single-agent environment.
- A person left alone in a maze is an example of the single-agent system.
- An environment involving more than one agent is a multi-agent environment.
- The game of football is multi-agent as it involves 11 players in each team.

5. Dynamic vs Static

- An environment that keeps constantly changing itself when the agent is up with some action is said to be dynamic.
- A roller coaster ride is dynamic as it is set in motion and the environment keeps changing every instant.
- An idle environment with no change in its state is called a static environment.
- An empty house is static as there's no change in the surroundings when an agent enters.

6. Discrete vs Continuous

- If an environment consists of a finite number of actions that can be deliberated in the environment to obtain the output, it is said to be a discrete environment.
- The game of chess is discrete as it has only a finite number of moves. The number of moves might vary with every game, but still, it's finite.
- The environment in which the actions are performed cannot be numbered i.e. is not discrete, is said to be continuous.
- Self-driving cars are an example of continuous environments as their actions are driving, parking, etc. which cannot be numbered.

7. Episodic vs Sequential

- In an **Episodic task environment**, each of the agent's actions is divided into atomic incidents or episodes. There is no dependency between current and previous incidents. In each incident, an agent receives input from the environment and then performs the corresponding action.
- **Example:** Consider an example of **Pick and Place robot**, which is used to detect defective parts from the conveyor belts. Here, every time robot(agent) will make the decision on the current part i.e. there is no dependency between current and previous decisions.
- In a **Sequential environment**, the previous decisions can affect all future decisions. The next action of the agent depends on what action he has taken previously and what action he is supposed to take in the future.
- **Example:**
 - **Checkers-** Where the previous move can affect all the following moves.

8. Known vs Unknown

- In a known environment, the output for all probable actions is given. Obviously, in case of unknown environment, for an agent to make a decision, it has to gain knowledge about how the environment works.

Question Bank:

1. What is Artificial Intelligence (AI)? Provide a formal definition.
2. Briefly describe the history and evolution of AI.
3. Differentiate between Artificial Narrow Intelligence (ANI), Artificial General Intelligence (AGI), and Artificial Super Intelligence (ASI) with examples.
4. What are the key concepts of production systems in AI?
5. Define an agent in the context of AI.
6. Explain the different types of environments in AI.
7. What are the characteristics of an intelligent agent?
8. Define rationality in AI. How does it influence decision-making in intelligent agents?
9. Describe the different types of agent with examples.